A CLASSICAL BLOCK CALCULATION: UNVERIFIED RESEARCH DRAFT

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ABSTRACT. This internal draft records a classical block quadratic-form criterion and its elementary proof. Its sole principal result is known. No original contribution, authorized original derivation or complete research-priority investigation has been supplied; formal research release remains unavailable.

1. KNOWN MATHEMATICS AND THE MISSING RESEARCH CONTRACT

The result below is supported by Boyd and Vandenberghe [1, Appendix A.5.5]. The purpose of the calculation is to make its hypotheses and mechanism inspectable, not to claim a new theorem.

Theorem 1.1 (Known classical result)**.** *Let $p, q \geq 1$, let $A \in \mathbb{R}^{p\times p}$ and $C \in \mathbb{R}^{q\times q}$ be symmetric, and let $B \in \mathbb{R}^{p\times q}$. Assume $A > 0$, where $H > 0$ means $v^T H v > 0$ for every nonzero real vector v of the appropriate size. Set*

$$M = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix}, \qquad S = C - B^T A^{-1} B.$$

Then $M > 0$ if and only if $S > 0$. Moreover, for every fixed $y \in \mathbb{R}^q$,

$$\min_{x\in\mathbb{R}^p} (x, y)^T M (x, y) = y^T S y, \qquad x_{\min} = -A^{-1} B y.$$

The minimizer is unique. This is a classical quadratic-form identity and criterion.

The contribution classification of Theorem 1.1 is *known*. There is no mathematical difference from the cited criterion being asserted as a contribution.

No candidate original principal is stated. A future research claim would need its own authorized provenance, closest verified prior results, precise comparison and complete proof before it could be evaluated.

2. MECHANISM AND COMPLETE PROOF OF THE KNOWN RESULT

For the fixed blocks, let

$$Q(x, y) = x^T A x + 2x^T B y + y^T C y, \qquad z = x + A^{-1} B y.$$

The inverse exists because $A > 0$. Since A^{-1} is symmetric,

$$z^T A z = x^T A x + 2x^T B y + y^T B^T A^{-1} B y.$$

Therefore

$$Q(x, y) = z^T A z + y^T S y, \qquad S = C - B^T A^{-1} B. \tag{2.1}$$

The map $(x, y) \mapsto (z, y)$ is invertible, with $x = z - A^{-1} B y$. It preserves whether the pair is nonzero.

If $S > 0$, the right side of (2.1) is positive for every nonzero pair, proving $M > 0$. Conversely, if $M > 0$, take any $y \neq 0$ and set $x = -A^{-1}By$. This gives $0 < Q(x, y) = y^T Sy$, hence $S > 0$.

For fixed y, $z^T Az \geq 0$, with equality exactly at $z = 0$. The unique minimizing x is consequently $-A^{-1}By$, and the minimum is $y^T Sy$. This closes the elementary proof of the known principal.

For $A = 2, B = C = 1$, one directly obtains

$$Q(x, y) = 2(x + y/2)^2 + y^2/2.$$

For $A = -1, B = 0, C = 1$, the identity still holds but gives $Q(x, y) = -x^2 + y^2$. Then $S = 1 > 0$ while M is indefinite and the fixed-y problem is unbounded below.

3. Unverified research responsibilities

The complete proof above concerns only a known theorem. It supplies no original-research full-proof record, because no original theorem is proposed in this draft.

The source check establishes what the cited classical result says. It is not an exhaustive prior-art search and cannot support a claim that a field has no earlier result of a proposed kind.

Authorization to prepare this internal demonstration does not provide provenance or authorization for an original contribution. Neither author rereading nor a nonauthor AI check changes that boundary.

The draft therefore retains three unsatisfied research conditions:

- an actual original claim with authorized provenance;
- dated, sufficiently broad prior-art work and exact prior comparisons;
- a complete proof and review of that original claim.

DRAFT_UNVERIFIED. This file may be inspected as an internal draft. It must not be relabelled, packaged or released as an accepted original research paper.

References

[1] Stephen Boyd and Lieven Vandenberghe. Convex optimization. Online source, 2004. published; controlling version: 2004; author-hosted PDF accessed 2026-10-12.